CNC Pancake Maker
Specifications Document

Johnathan Powell

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Specifications

1 SCOPE

- **General:** This document describes the design and verification requirements for the CNC Pancake Maker system. This system is a device that can be used with a standard electric griddle to create novelty pancakes from an image file. The device will process the image, generate a set of machine instructions, then follow those instructions to extrude pancake batter onto the griddle in correct patterns and with correct timing as to create the image.
- **Acronyms:**
    - CNC - Computer Numerical Control; describing a system or a process that involves the use of computers to control machine tools.

2 APPLICABLE DOCUMENTS

- Title 21 Code of Federal Requirements Part 174 - Indirect Food Additives: General (21 CFR 174)

3 ENGINEERING REQUIREMENTS

3.1 Item Description

The CNC Pancake Maker is designed to be used with an existing electric cooking griddle to create novelty pancakes from an image file. The system will consist of four parts: the pre-processing software, the interface software, the controller hardware, and the mechanical module (refer to Figure 1).

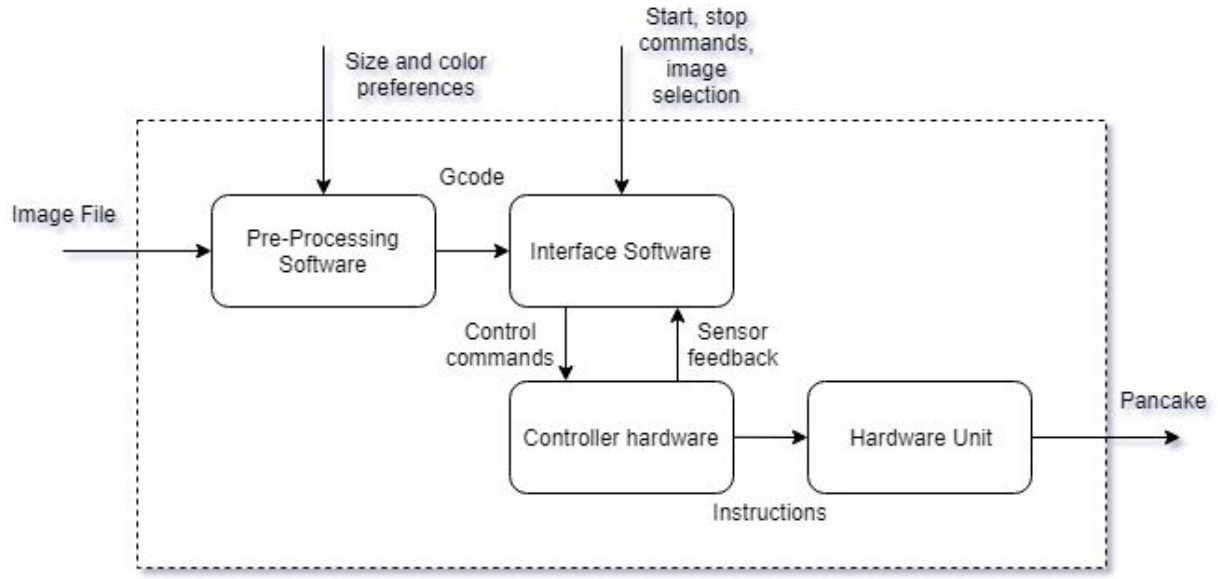


Figure 1: Modularized Functional Diagram

The system will take as input an image file, as well as size and color selections and start and stop commands. After any needed processing, the mechanical components of the system will extrude pancake batter to the griddle in order to produce the desired pattern or image, then after the batter has had time to set up properly, alert the user so that the user may complete the remainder of the process of cooking the pancake, including flipping and removal.
The non-functional parts of the system will consist of a power supply, and appropriate packaging for portability, ease of storage, and food-safe hardware.

The following sections describe the requirements for the interface, functional, and non-functional components of the system.

3.2 Interface Requirements

3.2.1 Physical

3.2.1.1 The system shall be able to print on a griddle from 3 to 4 inches in height.

3.2.1.2 The system shall have a print area of at least 8 by 8 inches.

3.2.1.3 Any portion of the system which extends under the griddle shall not exceed 1 inch in height.

3.2.2 Functional

3.2.2.1 The interface shall be available on both an external monitor or an integrated display.

3.2.2.2 The interface shall notify users when a pancake is complete with both visual and audible alerts, as configured by the user.

3.3 Functional Requirements

3.3.1 Pre-processing Software

3.3.1.1 The pre-processor shall accept image files in .png format.

3.3.1.2 The pre-processor shall accept image files from a USB flash drive.

3.3.1.3 The pre-processor shall allow the user to adjust the contrast and darkness thresholds for the image.

3.3.2 Interface Software

3.3.2.1 The interface software shall allow the user to select from any past images that were used.

3.3.2.2 The interface software shall provide the user with an image preview.

3.3.2.3 The interface software shall provide the user with start and stop prompts.

3.3.3 Control Hardware

3.3.3.1 The control hardware shall receive machine commands from the interface software and relay the correct instructions to the hardware unit.

3.3.3.2 The control hardware shall have any needed endstops and sensors needed for operation.

3.3.3.3 The control hardware shall be able to detect when the pancake batter is depleted, and pause operation.

3.3.4 Hardware Unit

3.3.4.1 The hardware unit shall not exceed 70 dBA under normal operation.

3.3.4.2 The hardware unit shall be able to hold at least as much pancake batter as is needed for a 8-inch diameter pancake.

3.4 Non-Functional Requirements

3.4.1 Electrical

3.4.1.1 The system shall be powered by a standard American 120 V, 60 Hz household outlet.

3.4.2 Physical

3.4.2.1 The system shall fit within an area no more than 20 inches by 20 inches by 20 inches when stored (when not in use).

3.4.3 Environmental

3.4.3.1 The operating temperature range of this system shall be 0 to 40° C.

3.4.3.2 Any components which come in contact with food (under normal operation) shall meet requirements outlined in 21 CFR 174 for an indirect food additive

3.4.3.3 Any components which come in contact with food (under normal operation) shall be removable for washing.

3.4.4 Economic

3.4.4.1 The system prototype shall have a one-time cost of under $250.

3.5 Other Requirements

3.5.1 Safety

3.5.1.1 The system shall not have any exposed conductors.

3.5.1.2 The system shall have all wires and cables correctly secured and protected.

3.5.1.3 The system shall have all electronics and electronic components enclosed in a manner sufficient to withstand minor splashes and spills.

3.5.1.4 The system shall have all pinch points clearly indicated with appropriate warning signage. A pinch point is defined as anywhere in the system where it is possible for a person or part of a person's body to be caught between moving or stationary parts.

3.5.2 Required equipment

3.5.2.1 Use of the system shall not require the use of a personal laptop or computer, other than to load images to the USB flashdrive.

4 VERIFICATION OF REQUIREMENTS

4.1 Item Description

The following sections describe any testing procedures needed to verify that the engineering requirements defined above have been met. Each test item corresponds to a requirement of the equivalent enumeration.

Note that most of these requirements can be verified by running a job from start to finish.

4.2 Interface Requirements

4.2.1 Physical

4.2.1.1 Verify that the system can be adjusted to extrude on a surface from 3 to 4 inches in height.

4.2.1.2 Use manual controls to verify that the machine has a printable area of at least 8 by 8 inches.

4.2.1.3 Measure that any portion of the system which extends under the griddle does not exceed 1 inch in height.

4.2.2 Functional

4.2.2.1 Verify that the interface can be viewed on an external display or on an integrated display.

4.2.2.2 Verify that visual and audible alerts are given when a job finishes, and that both can be disabled by the user.

4.3 Functional Requirements

4.3.1 Pre-Processing Software

4.3.1.1 Load an image in .png format and verify that it is correctly recognized.

4.3.1.2 Verify that image files are accepted from a USB flashdrive.

4.3.1.3 Verify that contrast and darkness thresholds can be configured.

4.3.2 Interface Software

4.3.2.1 Verify that the user can select from a list of previously processed images.

4.3.2.2 Verify that an image preview is provided.

4.3.2.3 Inspect the system to verify that start and stop prompts are present.

4.3.3 Control Hardware

4.3.3.1 Verify that the control hardware responds when start and stop prompts on interface software are used.

4.3.3.2 Verify that endstops and sensors are present and correctly connected.

4.3.3.3 Verify that operation stops when pancake batter is depleted.

4.3.4 Mechanical Unit

4.3.4.1 Use a decibel meter to verify that system does not exceed 70 dBA when operating.

4.3.4.2 Print a pancake 8 inches in diameter and verify that there is sufficient pancake batter.

4.4 Non-Functional Requirements

4.4.1 Electrical

4.4.1.1 Verify that the system is properly equipped to be powered by a standard American household outlet, and that it operates correctly when so powered.

4.4.2 Physical

4.4.2.1 Verify that the system can be stored in an area no more than 20 inches by 20 inches by 20 inches.

4.4.3 Environmental

4.4.3.1 Verify from material datasheets or documentation that all materials used are safe for use from 0 to 40° C.

4.4.3.2 Inspect the system and verify datasheets to ensure that all components which come into contact with food (under normal operation) meet requirements outlined in 21 CFR 174 for an indirect food additive.

4.4.3.3 Inspect the system and verify that all components which come into contact with food (under normal operation) are easily removable.

4.4.4 Economic

4.4.4.1 Review the bill of materials and/or purchase records to verify that the cost of the system prototype does not exceed $250.

4.5 Other Requirements

4.5.1 Safety

4.5.1.1 Inspect the system to verify that there are no exposed conductors so that the user does not come in contact with them when the system is in use.

4.5.1.2 Inspect the system to verify that all pinch points are clearly marked with clear and understandable warning signs.

4.5.2 Required Equipment

4.5.2.1 Run a job from start to stop to verify that a computer is not needed.

Paragraph Number	Tester's Name	Pass/Fail	Date